

1 Solution — GIAM 5.3

Problem 10

1.1 Problem

Give an inductive proof that $\forall n \geq 3 \in \mathbb{N}, 3n^2 + 3n + 1 < 2n^3$.

1.2 Answer

$$D(k+1) = D(k) + 6k^2 - 4, \quad D(n) = 2n^3 - (3n^2 + 3n + 1)$$

The gap D starts positive at the basis, $D(3) = 54 - 37 = 17$, and only grows, since $6k^2 - 4 > 0$ for $k \geq 1$. Verified for every n from 3 to 5000; the inequality fails at $n = 0, 1, 2$.

1.3 The Proof

Write $L(n) = 3n^2 + 3n + 1$ and $R(n) = 2n^3$.

Theorem. For every natural number $n \geq 3$, $L(n) < R(n)$.

Proof. By induction on n , starting at 3.

Basis. At $n = 3$, $L(3) = 27 + 9 + 1 = 37$ and $R(3) = 54$, and $37 < 54$.

Inductive step. Let $k \geq 3$ be a particular but arbitrarily chosen natural number and suppose $L(k) < R(k)$. Expanding both sides at $k + 1$,

$$\begin{aligned} L(k+1) &= 3(k+1)^2 + 3(k+1) + 1 = L(k) + 6k + 6, \\ R(k+1) &= 2(k+1)^3 = R(k) + 6k^2 + 6k + 2. \end{aligned}$$

The right side gains more than the left, because

$$(6k^2 + 6k + 2) - (6k + 6) = 6k^2 - 4 > 0 \quad \text{for } k \geq 1.$$

Combining this with the hypothesis,

$$\begin{aligned} L(k+1) &= L(k) + 6k + 6 \\ &< R(k) + 6k + 6 \\ &< R(k) + 6k^2 + 6k + 2 = R(k+1). \end{aligned}$$

So $L(k+1) < R(k+1)$.

Basis and step together give the result for every $n \geq 3$. ■

The step uses only $k \geq 1$, not $k \geq 3$: what restricts the theorem to $n \geq 3$ is the basis, not the step. This is the pattern of inequality proofs by induction: compare the *increments* of the two sides, and show the larger side also grows faster.

1.4 The Picture

The bars show the crossover; the second panel shows the cleaner form the inequality takes.

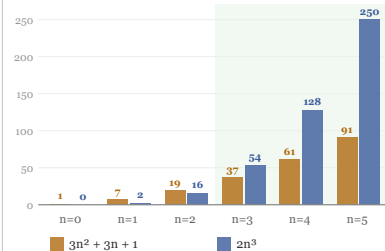
THE GAP GROWS AT EVERY STEP

$$D(k+1) = D(k) + 6k^2 - 4, \text{ where } D(n) = 2n^3 - (3n^2 + 3n + 1)$$

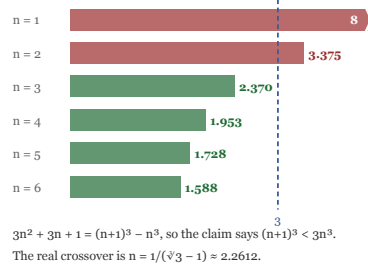
$D(3) = 17 > 0$, and $6k^2 - 4 > 0$ for $k \geq 1$, so $D(n) > 0$ for every $n \geq 3$. Verified for n up to 5000.

the cubic overtakes the quadratic at $n = 3$

true from the basis $n = 3$ on



equivalently $(1 + 1/n)^3 < 3$



WHY THE BASIS IS 3

The claim fails at $n = 0, 1, 2$ ($1 < 0$, $7 < 2$, $19 < 16$ are all false) and holds at $n = 3$ ($37 < 54$).

The step itself is valid from $k = 1$, so the basis, not the step, is what limits the theorem.

THE INCREMENTS TELL THE STORY

Each step adds $2(3k^2 + 3k + 1)$ to the right side, twice the left side's current value, but only $6(k+1)$ to the left. Once ahead, the cubic pulls away: the ratio of left to right tends to 0.

Figure 1.1: A diagram in four parts. The top band gives the step identity for the gap D of n . The left panel compares three n squared plus three n plus one with two n cubed for n from zero to five, the cubic taking the lead at n equal to three. The right panel shows one plus one over n , cubed, against a dashed line at three, crossing below it at n equal to three. Two bands explain why the basis is three and compare the increments of the two sides.

1.5 Remark — A Hidden Difference of Cubes

The left side is not an arbitrary quadratic. By the binomial theorem,

$$(n+1)^3 - n^3 = 3n^2 + 3n + 1,$$

so the inequality says $(n+1)^3 - n^3 < 2n^3$, that is, $(n+1)^3 < 3n^3$. Dividing by n^3 ,

$$\left(1 + \frac{1}{n}\right)^3 < 3 \iff 1 + \frac{1}{n} < \sqrt[3]{3} \iff n > \frac{1}{\sqrt[3]{3} - 1} \approx 2.2612.$$

The cubic $2x^3 - 3x^2 - 3x - 1$ has exactly this one real root, so the inequality holds for real x above 2.2612 and fails below it. The first natural number past the root is 3, which is why the problem starts there. In the figure's right panel the values 8, 3.375, 2.370, 1.953, ... of the left-hand side fall below 3 exactly at $n = 3$.

This gives a proof without induction: for $n \geq 3$, $(1 + 1/n)^3 \leq (4/3)^3 = 64/27 < 3$.

1.6 Remark — The Basis Is Forced

The claim fails at each of the earlier candidates:

n	$3n^2 + 3n + 1$	$2n^3$	holds?
0	1	0	no
1	7	2	no
2	19	16	no
3	37	54	yes

Yet the inductive step is valid from $k = 1$ onward. So a proof starting at $n = 1$ would have a sound step and a false basis, the situation of 5.3 Problem 9 but now only for a finite stretch. Starting at 3 is not a matter of taste: it is the first point where the basis is

true. In general, induction may start from any natural number b , and it then proves the claim for exactly the $n \geq b$.

1.7 Remark — The Increments Tell the Story

The two increments are themselves linked. Since $L(k) = (k+1)^3 - k^3$,

$$R(k+1) - R(k) = 2((k+1)^3 - k^3) = 2L(k),$$

while $L(k+1) - L(k) = 6(k+1)$; both identities were checked for k up to 1000. The right side grows by twice the current value of the left side, which quickly dwarfs the left side's linear increment. Once the cubic is ahead it pulls away: the ratio $L(n)/R(n)$ falls from 0.685 at $n = 3$ to 0.015 at $n = 100$, and tends to 0.

The same identity drives the power sums of 5.2: summing $(k+1)^3 - k^3 = 3k^2 + 3k + 1$ from $k = 0$ to $n-1$ telescopes to n^3 , which is how the formula for $\sum k^2$ is usually derived.